

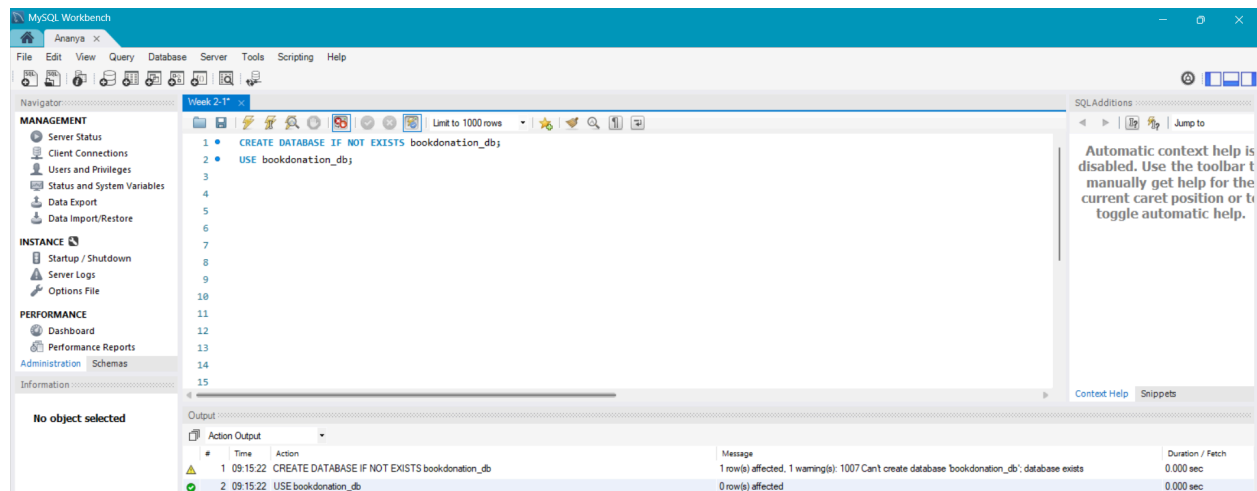
Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

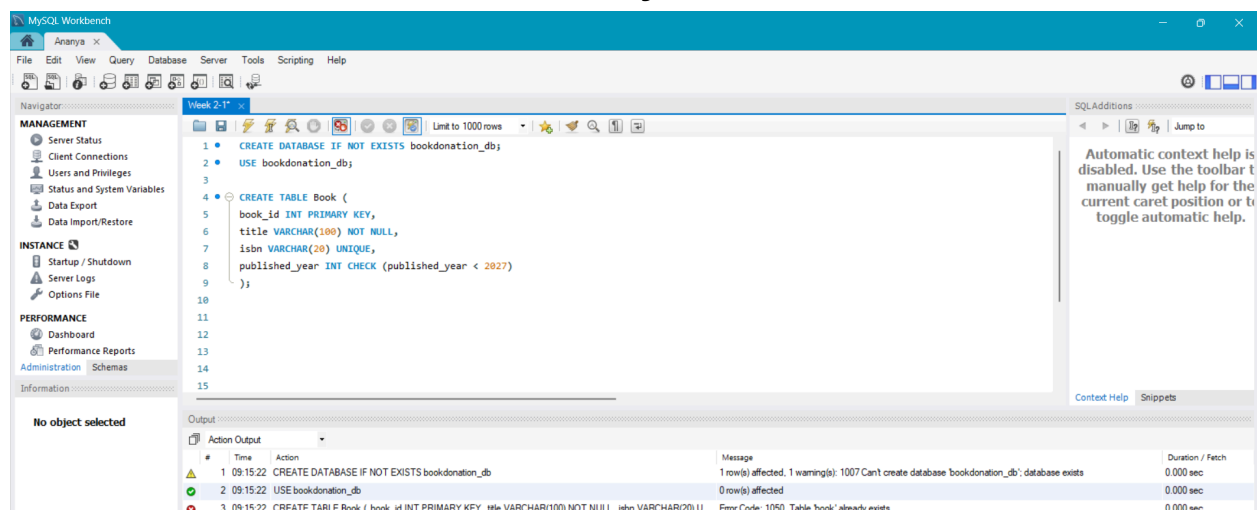
Week-2

T.Ananya
2520030039

Query-1:



Query-2:



Query-3:

MySQL Workbench

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports

Administration Schemas

Information

No object selected

SQL Editor

```

1 • CREATE DATABASE IF NOT EXISTS bookdonation_db;
2 • USE bookdonation_db;
3
4 • CREATE TABLE Book (
5   book_id INT PRIMARY KEY,
6   title VARCHAR(100) NOT NULL,
7   isbn VARCHAR(20) UNIQUE,
8   published_year INT CHECK (published_year < 2027)
9 );
10
11 • INSERT INTO Book (book_id, title, isbn, published_year) VALUES
12   (1, 'The Great Gatsby', '9780743273565', 1925),
13   (2, 'To Kill a Mockingbird', '9780061120084', 1960),
14   (3, '1984', '9780451524935', 1949);
15

```

Output

#	Time	Action	Message	Duration / Fetch
1	09:15:22	CREATE DATABASE IF NOT EXISTS bookdonation_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'bookdonation_db': database exists	0.000 sec
2	09:15:22	USE bookdonation_db	0 row(s) affected	0.000 sec
3	09:15:22	CREATE TABLE Book (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(20) U...	Error Code: 1050. Table 'book' already exists	0.000 sec
4	09:15:22	INSERT INTO Book (book_id, title, isbn, published_year) VALUES (1, 'The Great Gatsby', '9780743273565', 1925), (2, 'To Kill a Mockingbird', '9780061120084', 1960), (3, '1984', '9780451524935', 1949);	Error Code: 1062. Duplicate entry '1' for key 'book.PRIMARY'	0.000 sec

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Query-4:

MySQL Workbench

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports

Administration Schemas

Information

No object selected

SQL Editor

```

12   (1, 'The Great Gatsby', '9780743273565', 1925),
13   (2, 'To Kill a Mockingbird', '9780061120084', 1960),
14   (3, '1984', '9780451524935', 1949);
15
16 • SELECT * FROM Book;

```

Result Grid

book_id	title	isbn	published_year
1	The Great Gatsby	9780743273565	1925
2	To Kill a Mockingbird	9780061120084	1960
3	1984	9780451524935	1949

Output

#	Time	Action	Message	Duration / Fetch
1	09:15:22	CREATE DATABASE IF NOT EXISTS bookdonation_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'bookdonation_db': database exists	0.000 sec
2	09:15:22	USE bookdonation_db	0 row(s) affected	0.000 sec
3	09:15:22	CREATE TABLE Book (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(20) U...	Error Code: 1050. Table 'book' already exists	0.000 sec
4	09:15:22	INSERT INTO Book (book_id, title, isbn, published_year) VALUES (1, 'The Great Gatsby', '9780743273565', 1925), (2, 'To Kill a Mockingbird', '9780061120084', 1960), (3, '1984', '9780451524935', 1949);	Error Code: 1062. Duplicate entry '1' for key 'book.PRIMARY'	0.000 sec
5	09:15:22	SELECT * FROM Book LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Query-5:

MySQL Workbench interface showing the execution of Query-5. The SQL editor contains the following script:

```
14 (3, '1984', '9780451524935', 1949);
15
16 * SELECT * FROM Book;
17
18 * CREATE TABLE Members (
19     member_id INT PRIMARY KEY,
20     full_name VARCHAR(100),
21     email VARCHAR(100) UNIQUE
22 );
23 * INSERT INTO Members (member_id, full_name, email) VALUES
24     (101, 'John Smith', 'john.smith@email.com'),
25     (102, 'Emma Wilson', 'emma.wilson@email.com'),
26     (103, 'Michael Brown', 'michael.brown@email.com');
27 * SELECT * FROM Members;
```

The output pane displays the execution log:

#	Time	Action	Message	Duration / Fetch
1	09:18:53	CREATE DATABASE IF NOT EXISTS bookdonation_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'bookdonation_db'; database exists	0.000 sec
2	09:18:53	USE bookdonation_db	0 row(s) affected	0.000 sec
3	09:18:53	CREATE TABLE Book (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(20) UNIQUE, published_year INT)	Error Code: 1050. Table 'book' already exists	0.000 sec
4	09:18:53	INSERT INTO Book (book_id, title, isbn, published_year) VALUES (1, 'The Great Gatsby', '9780743273565', 1925), (2, 'To Kill a Mockingbird', '9780061120084', 1960), (3, '1984', '9780451524935', 1949)	Error Code: 1062. Duplicate entry '1' for key 'book.PRIMARY'	0.000 sec
5	09:18:53	SELECT * FROM Book LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
6	09:18:53	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100) UNIQUE)	0 row(s) affected	0.031 sec
7	09:18:53	INSERT INTO Members (member_id, full_name, email) VALUES (101, 'John Smith', 'john.smith@email.com'), (102, 'Emma Wilson', 'emma.wilson@email.com'), (103, 'Michael Brown', 'michael.brown@email.com')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
8	09:18:53	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Query-6:

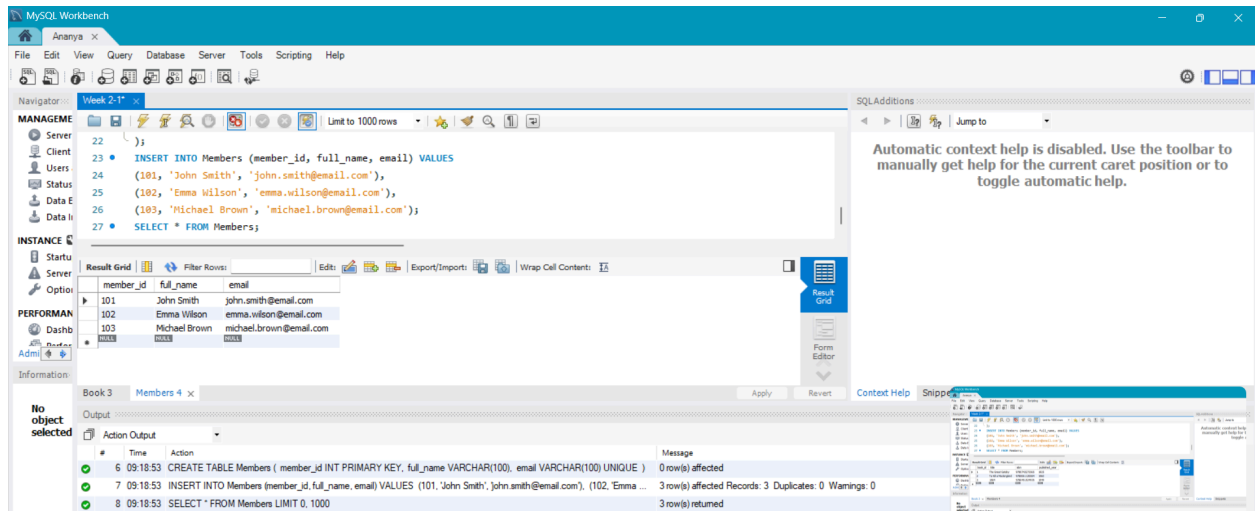
MySQL Workbench interface showing the execution of Query-6. The SQL editor contains the following script:

```
22 *
23 * INSERT INTO Members (member_id, full_name, email) VALUES
24     (101, 'John Smith', 'john.smith@email.com'),
25     (102, 'Emma Wilson', 'emma.wilson@email.com'),
26     (103, 'Michael Brown', 'michael.brown@email.com');
27 * SELECT * FROM Members;
```

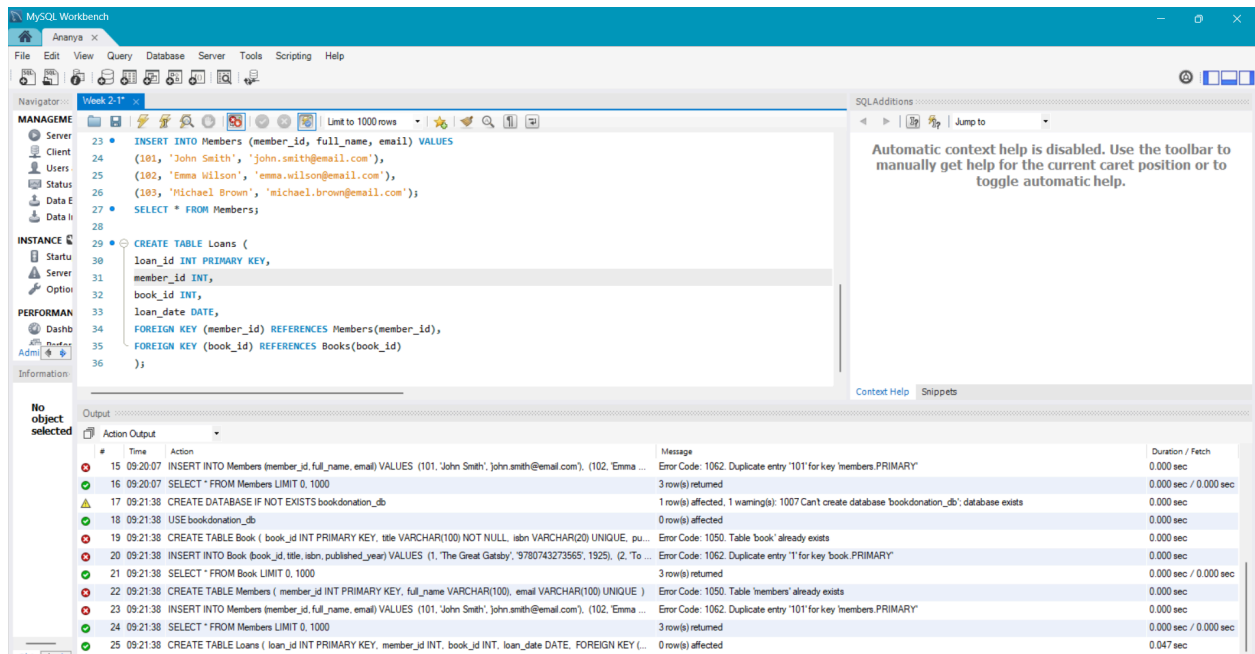
The output pane displays the execution log:

#	Time	Action	Message	Duration / Fetch
6	09:18:53	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100) UNIQUE)	0 row(s) affected	0.031 sec
7	09:18:53	INSERT INTO Members (member_id, full_name, email) VALUES (101, 'John Smith', 'john.smith@email.com'), (102, 'Emma Wilson', 'emma.wilson@email.com'), (103, 'Michael Brown', 'michael.brown@email.com')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
8	09:18:53	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

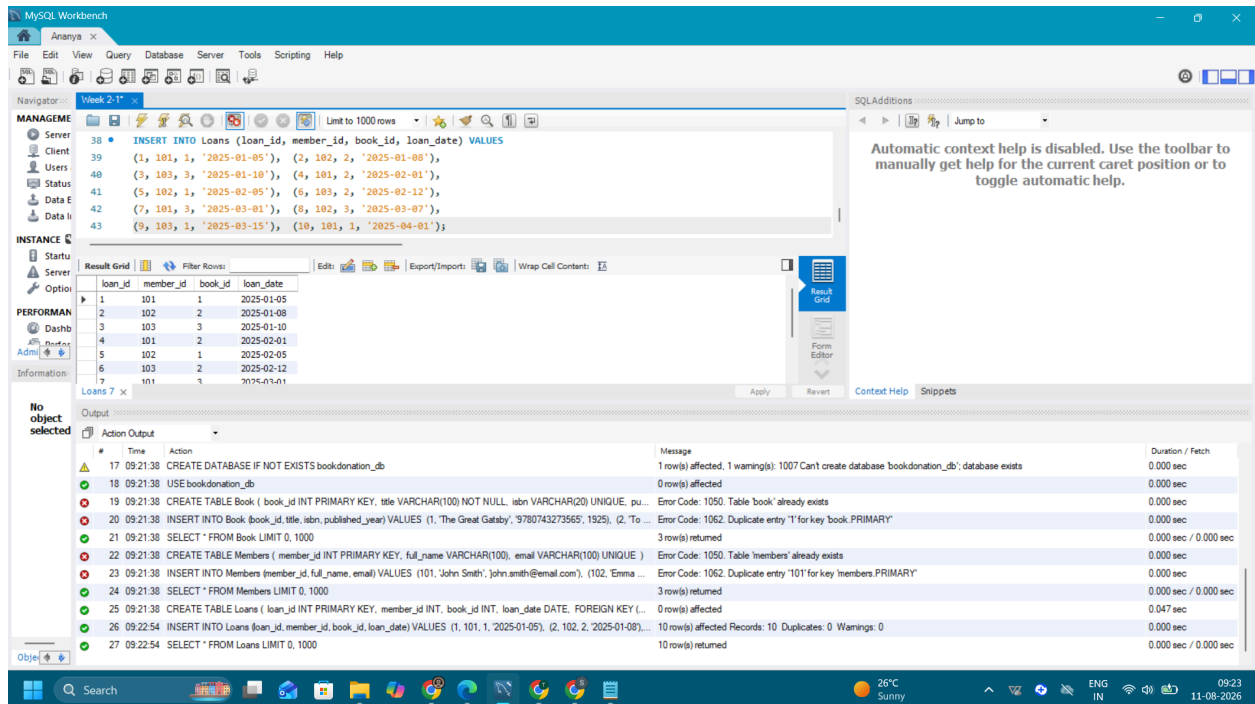
Query-7:



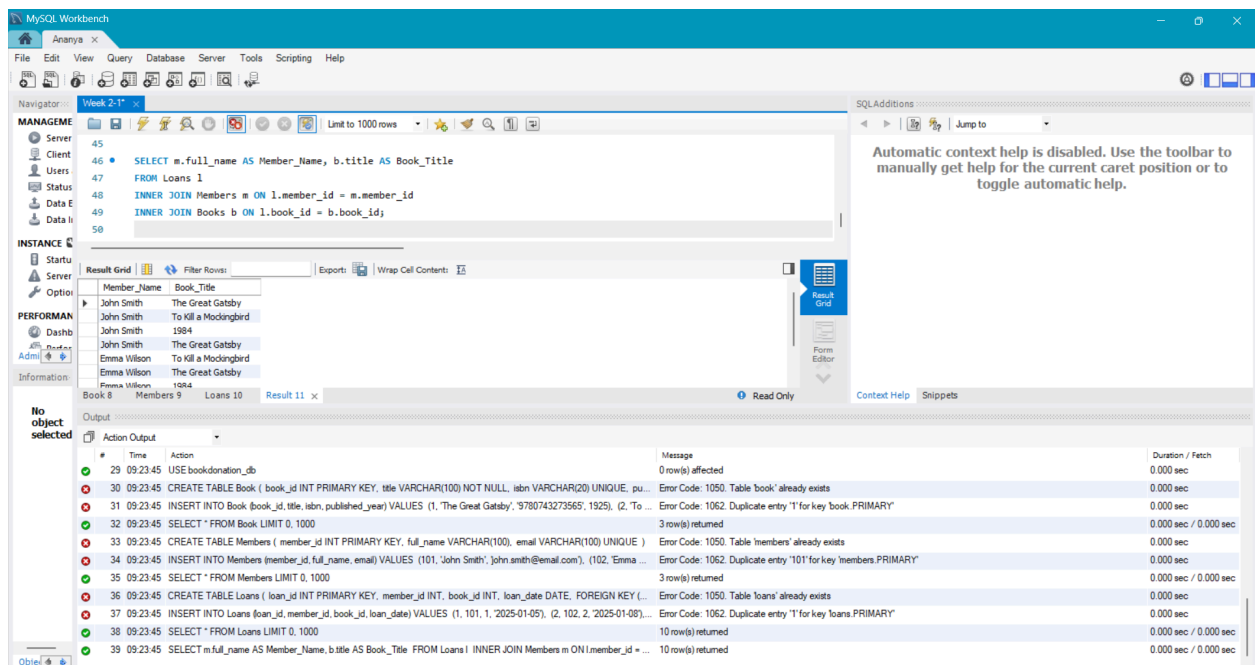
Query-8:



Query-9:



Query-10:



Query-11:

MySQL Workbench

Navigator: Week 2-1*

MANAGE

Server Client Users Status Data E Data li

INSTANCE

Startu Server Option

PERFORMAN

Dashb Admin

Information

SQL

```

49 INNER JOIN Books b ON l.book_id = b.book_id;
50
51 SELECT published_year, COUNT(book_id) AS Total_Books
52 FROM Books
53 GROUP BY published_year
54 ORDER BY published_year;

```

Result Grid

published_year	Total_Books
1925	1
1949	1
1960	1

Output

#	Time	Action	Message	Duration / Fetch
42	09:24:58	CREATE TABLE Book (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(20) UNIQUE, pu...	Error Code: 1050. Table 'book' already exists	0.000 sec
43	09:24:58	INSERT INTO Book (book_id, title, isbn, published_year) VALUES (1, 'The Great Gatsby', '9780743273565', 1925), (2, To ...	Error Code: 1062. Duplicate entry '1' for key 'book PRIMARY'	0.000 sec
44	09:24:58	SELECT * FROM Book LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
45	09:24:58	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100) UNIQUE)	Error Code: 1050. Table 'members' already exists	0.016 sec
46	09:24:58	INSERT INTO Members (member_id, full_name, email) VALUES (101, 'John Smith', 'john.smith@email.com'), (102, 'Emma ...	Error Code: 1062. Duplicate entry '101' for key 'members PRIMARY'	0.000 sec
47	09:24:58	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
48	09:24:58	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, FOREIGN KEY (...	Error Code: 1050. Table 'loans' already exists	0.000 sec
49	09:24:58	INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES (1, 101, 1, 2025-01-05), (2, 102, 2, 2025-01-08)...	Error Code: 1062. Duplicate entry '1' for key 'loans PRIMARY'	0.000 sec
50	09:24:58	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
51	09:24:58	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m ON l.member_id = ...	10 row(s) returned	0.000 sec / 0.000 sec
52	09:24:58	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year ORDER BY publi...	3 row(s) returned	0.000 sec / 0.000 sec

Query-12:

MySQL Workbench

Navigator: Week 2-1*

MANAGE

Server Client Users Status Data E Data li

INSTANCE

Startu Server Option

PERFORMAN

Dashb Admin

Information

SQL

```

59 donor_name VARCHAR(100),
60 donation_date DATE,
61 FOREIGN KEY (book_id) REFERENCES Books(book_id)
62 );
63
64 START TRANSACTION;
65
66 INSERT INTO Books (book_id, title, isbn, published_year)
67 VALUES (4, 'Animal Farm', '9780451526342', 1945);
68
69 INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
70 VALUES (1, 4, 'Raj Kumar', CURDATE());
71
72 COMMIT;

```

Output

#	Time	Action	Message	Duration / Fetch
48	09:24:58	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, FOREIGN KEY (...	Error Code: 1050. Table 'loans' already exists	0.000 sec
49	09:24:58	INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES (1, 101, 1, 2025-01-05), (2, 102, 2, 2025-01-08)...	Error Code: 1062. Duplicate entry '1' for key 'loans PRIMARY'	0.000 sec
50	09:24:58	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
51	09:24:58	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m ON l.member_id = ...	10 row(s) returned	0.000 sec / 0.000 sec
52	09:24:58	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year ORDER BY publi...	3 row(s) returned	0.000 sec / 0.000 sec
53	09:25:34	COMMIT	0 row(s) affected	0.000 sec
54	09:25:41	COMMIT	0 row(s) affected	0.000 sec
55	09:25:43	START TRANSACTION	0 row(s) affected	0.000 sec
56	09:25:43	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (4, 'Animal Farm', '9780451526342', 1945)	1 row(s) affected	0.000 sec
57	09:25:43	INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date) VALUES (1, 4, 'Raj Kumar', CURDAT...	Error Code: 1146. Table 'bookdonation_db_donation_history' doesn't exist	0.000 sec
58	09:25:43	COMMIT	0 row(s) affected	0.000 sec

Query-13:

MySQL Workbench

Ananya x

File Edit View Query Database Server Tools Scripting Help

Navigator: Week 2-1

MANAGEMENT

- Server
- Client
- Users
- Status
- Data E
- Data i

INSTANCE

- Startu
- Server
- Optio

PERFORMANCE

- Dashb
- Admi
- Information

SQL Editor

```
70 VALUES (1, 4, 'Raj Kumar', CURDATE());
71
72 COMMIT;
73
74 CREATE INDEX idx_books_isbn ON Books(isbn);
75 SELECT * FROM Books WHERE isbn = '9780451524935';
```

Result Grid

book_id	title	isbn	published_year
3	1984	9780451524935	1949

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Books 17 x

Output

Action Output

Time	Action	Message	Duration / Fetch
50 09:24:50	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
51 09:24:58	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m ON l.member_id = ...	10 row(s) returned	0.000 sec / 0.000 sec
52 09:24:58	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year ORDER BY publis...	3 row(s) returned	0.000 sec / 0.000 sec
53 09:25:34	COMMIT	0 row(s) affected	0.000 sec
54 09:25:41	COMMIT	0 row(s) affected	0.000 sec
55 09:25:43	START TRANSACTION	0 row(s) affected	0.000 sec
56 09:25:43	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (4, 'Animal Farm', '9780451526342', 1945)	1 row(s) affected	0.000 sec
57 09:25:43	INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date) VALUES (1, 4, 'Raj Kumar', CURDAT...	Error Code: 1146. Table 'bookdonation_db.donation_history' doesn't exist	0.000 sec
58 09:25:43	COMMIT	0 row(s) affected	0.000 sec
59 09:26:21	CREATE INDEX idx_books_isbn ON Books(isbn)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
60 09:26:26	SELECT * FROM Books WHERE isbn = '9780451524935' LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

Object